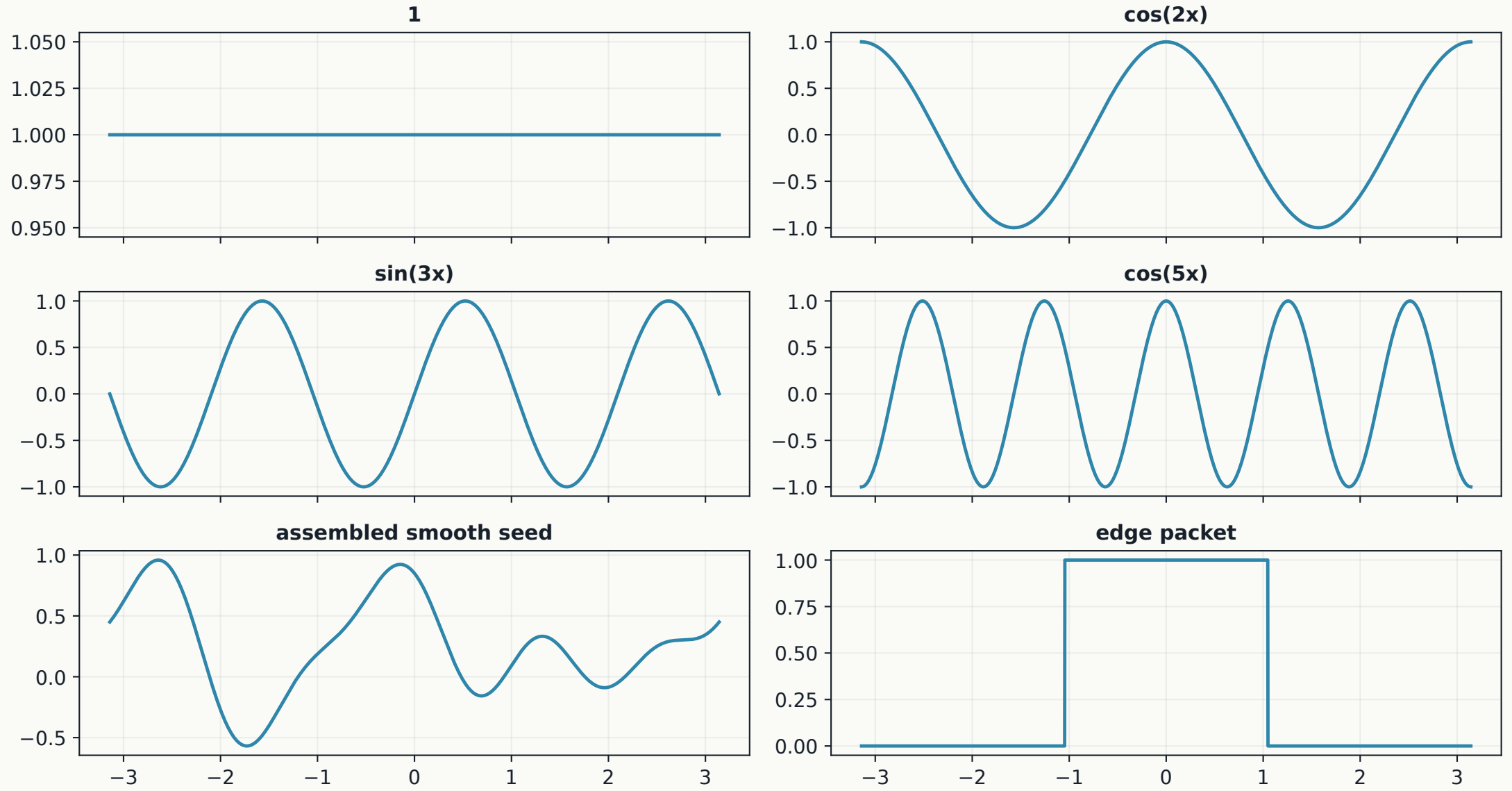Primitive atoms and the two signal packets